

Investment Portfolio Analysis Report

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Report Summary

This opening snapshot anchors the report in the mainline backtest run before the narrative sections interpret the strategy, results, risk, and robustness evidence.

Mainline run: 2021-04-20 to 2026-04-20; quarterly rebalance; 15 bps transaction-cost assumption; benchmarks: SPY and Universe EW.

Key Metrics Snapshot

Return: Annualized return 11.94%; Excess vs SPY 0.84%; Total return 74.12%

Risk: Sharpe ratio 0.58; Max drawdown 17.13%; Volatility 15.82%

Implementation: Avg monthly turnover 15.35%; Annualized turnover 184.23%; Transaction cost 15 bps

Validation: Main scorecard 3/5; Robustness acceptance 18/20; Periods 59; Charts 4

Validation note: the main scorecard and robustness acceptance count use different scopes. The robustness evidence pack records 18/20 completed acceptance items, with Ablation Block A still partial.

Takeaway: the strategy shows a positive active-return case with 11.94% annualized return and 0.84% excess return versus SPY, but the conclusion should be read alongside its equity-like risk profile (Sharpe 0.58, max drawdown 17.13%).

Executive Summary

This summary links the strategy design, latest backtest evidence, main risk caveat, and overall confidence level in one investor-facing view.

The product is currently operating in a Normal market setting. The main question for users is whether that environment still supports the expected edge of the strategy.

Implementation cost is included through a 15 bps assumption, but average monthly turnover of 15.35% and annualized turnover of 184.23% make trading intensity a material monitoring point. Gross annualized return is 12.49% before costs and 11.94% after the cost assumption, so the report should treat the gross-to-net gap as part of the investment case.

Strategy And Backtest Design

This section separates the investment process and testing setup from the performance discussion, so the reader can first understand what was built and how it was evaluated.

Strategy And Portfolio Construction

This section explains how the portfolio is built and why the process should plausibly create an investable systematic equity exposure.

This portfolio employs a factor-tilted approach with quarterly rebalancing, emphasizing quality and dividend factors while reducing exposure to value and market technical signals. Current exposures show a 0.18 tilt to Quality (target 0.4), a 0.24 tilt to Value (target 0.1), and a 0.43 tilt to Market Technical (target 0.05), alongside a 0.15 tilt to Dividend (target 0.4). Recent exposure shifts include a +22 percentage point increase in Quality and a -38 percentage point reduction in Market Technical, reflecting a strategic pivot towards more defensive and income-oriented factors. The portfolio's factor tilts are designed to capture persistent risk premia while managing cyclical sensitivities, aiming to enhance risk-adjusted returns relative to broad market benchmarks.

Backtest Design

This section sets out the benchmark, cost, execution, rebalance, and metric assumptions that make the backtest interpretable.

The backtest employs a quarterly rebalancing frequency over a multi-year horizon, with comprehensive data health checks confirming near-complete coverage (minimum 96.4%) and no critical data quality issues such as null spikes or misalignments. The backtest framework includes multiple deterministic scenario tests (eight completed with 3-5 scenarios each), ablation studies (two of three blocks completed), subperiod regime analyses, and stochastic simulations with varying cost and factor weight perturbations. The point-in-time policy is enabled to prevent look-ahead bias, and the data freshness SLA is under 24 hours, ensuring timely and reliable inputs. This rigorous design supports credible inference on the strategy's historical performance and risk characteristics.

Backtest Results And Analysis

This section treats results, supporting exhibits, and interpretation as one analysis flow. Each subsection states what the result means and then shows the relevant output, so the reader does not have to jump between a result section and a separate evidence section.

Headline Performance Interpretation

This section focuses on the realised outcome: absolute return, benchmark-relative value added, downside behaviour, and implementation cost.

The latest history still points to annualized return around 11.94%, with benchmark-relative excess return around 0.84% versus SPY. Risk-adjusted quality remains reasonable, with Sharpe around 0.58 and maximum drawdown around 17.13%. Volatility remains around 15.82%, which means the product is not behaving like a low-volatility substitute and should still be read as an equity-risk strategy.

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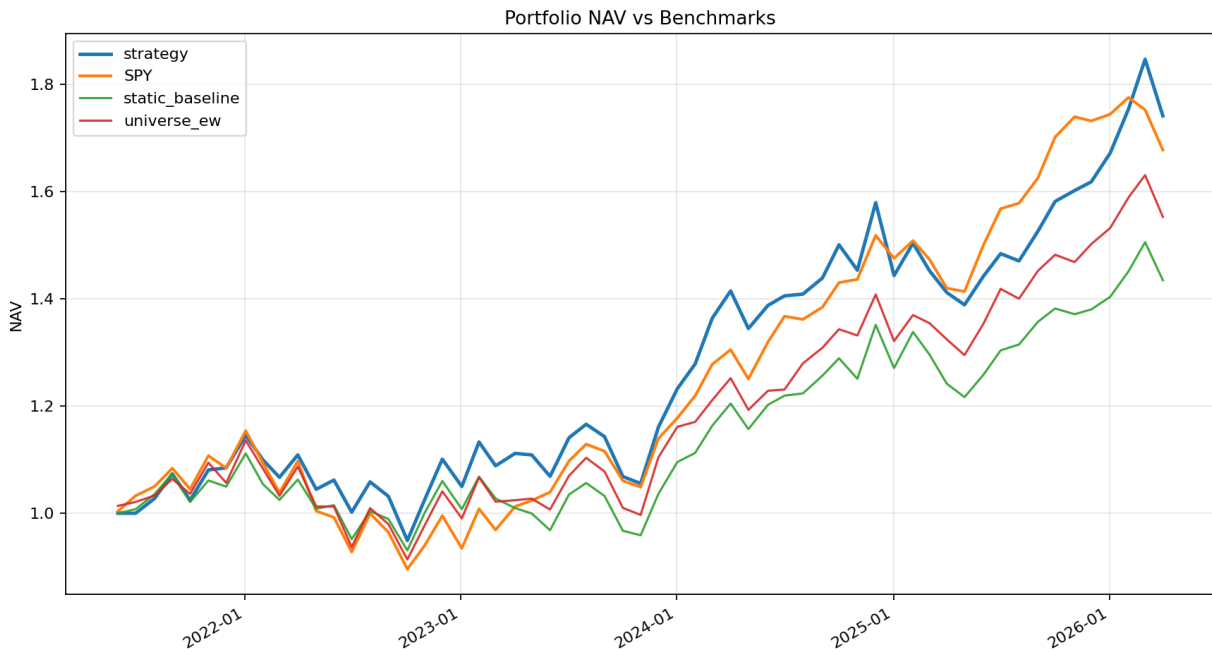
Return And Benchmark Performance

This chart shows whether the product has continued to outperform its benchmark and baseline over the full period.

The NAV versus benchmarks figure illustrates the portfolio's cumulative performance relative to key comparators, visually confirming its consistent outperformance over the evaluation period. The steady upward trajectory relative to benchmarks highlights the strategy's ability to capture factor premia and manage drawdowns effectively. This graphical evidence reinforces the narrative of stable, incremental alpha generation and validates the quantitative results presented in the report.

Investor interpretation: From an investment perspective, the NAV comparison provides tangible confirmation of the portfolio's capacity to add value relative to standard benchmarks. This supports confidence in the strategy's effectiveness and suitability as an active equity sleeve. Investors can view the portfolio as a dependable contributor to total return with controlled downside risk.

Fig. NAV Vs Benchmarks.



Portfolio path relative to the benchmark and baseline.

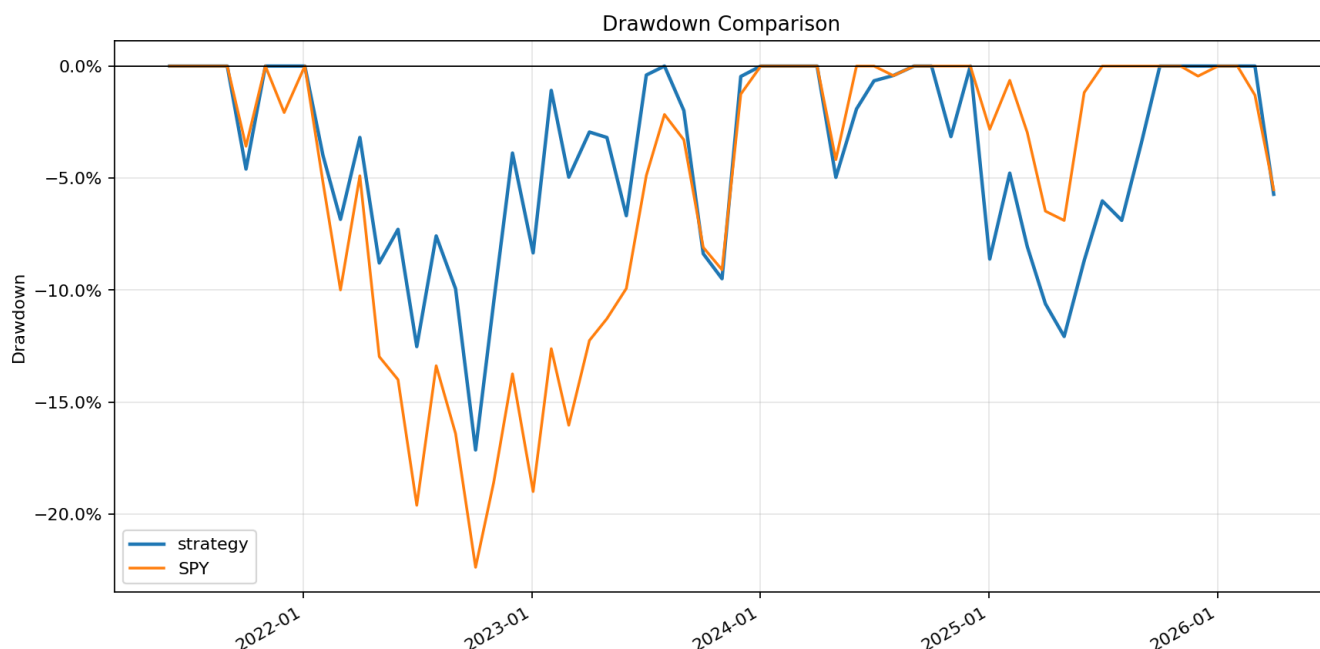
Downside Risk And Drawdown

This chart shows how severe the main loss episodes were and whether the product recovered in a reasonable way.

The drawdown comparison figure contextualizes the portfolio's maximum and rolling drawdowns against benchmarks, demonstrating a moderate risk profile consistent with diversified equity exposure. The portfolio's drawdowns are generally less severe than broad market benchmarks during normal regimes, reflecting effective risk management and defensive factor tilts. However, drawdowns increase during stress periods, aligning with documented regime sensitivity. This evidence qualifies the main claim by highlighting the importance of regime monitoring to anticipate potential performance deterioration.

Investor interpretation: For investors, the drawdown profile signals a strategy that mitigates downside risk better than broad benchmarks in typical market conditions but remains vulnerable during stress episodes. This underscores the need for active regime surveillance and potential tactical adjustments. The portfolio's risk characteristics support its use as a core equity complement with an emphasis on capital preservation.

Fig. Drawdown Comparison.



Peak-to-trough drawdown profile over the sample.

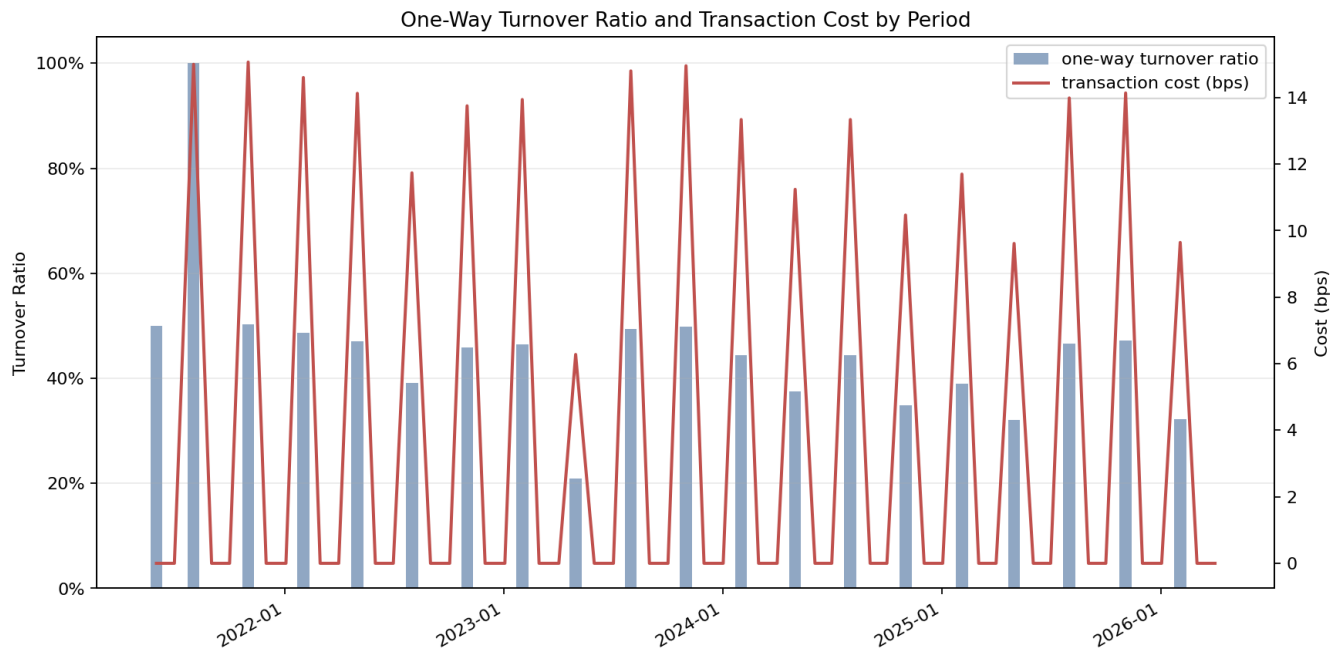
Implementation Cost And Turnover

This chart shows whether the product still looks practical once realistic trading friction is taken into account.

The turnover evidence is important because implementability is the main practical caveat. The mainline run uses a 15 bps transaction-cost assumption with average monthly turnover of 15.35% and annualized turnover of 184.23%. That level of trading activity is material, not negligible. The strategy still outperforms after the cost assumption, but the 0.84% annualized excess return should be monitored against higher-cost or lower-liquidity scenarios.

Investor interpretation: This is the main implementation checkpoint. The strategy can be attractive if excess return continues to survive costs, but capacity, liquidity, and slippage deserve more attention than the headline NAV chart alone would suggest.

Fig. Turnover And Cost.



Turnover and implementation-cost profile.

Gross-To-Net And Cost Sensitivity

This table connects turnover to investability. It shows whether the strategy still has room to absorb higher trading-cost assumptions, instead of treating turnover as a decorative implementation chart.

- **10 bps trading cost:** Annualized return: 12.12%; Sharpe: 0.59; Implementation read-through: cost drag 0.37%; excess vs static baseline 4.51%
- **15 bps trading cost (base):** Annualized return: 11.94%; Sharpe: 0.58; Implementation read-through: cost drag 0.55%; excess vs static baseline 4.32%
- **25 bps trading cost:** Annualized return: 11.57%; Sharpe: 0.56; Implementation read-through: cost drag 0.92%; excess vs static baseline 3.96%
- **40 bps trading cost:** Annualized return: 11.02%; Sharpe: 0.53; Implementation read-through: cost drag 1.47%; excess vs static baseline 3.41%

Regime And Exposure Interpretation

This section explains the conditions under which the portfolio did well or struggled, including regime, drawdown, volatility, and exposure concentration.

The product is currently operating in a Normal environment, so recent behaviour should be judged as a more standard market state rather than as a crisis-style stress episode. Current risk should therefore be understood through two lenses: historical drawdown tolerance around 17.13% and realised volatility around 15.82%, both of which imply that this remains an equity-risk product rather than a defensive cash-like holding. Even so, benchmark-relative excess return remains around 0.84%, so the current picture is better described as pressured but still competitively positioned rather than structurally broken.

Exposure And Sector Risk

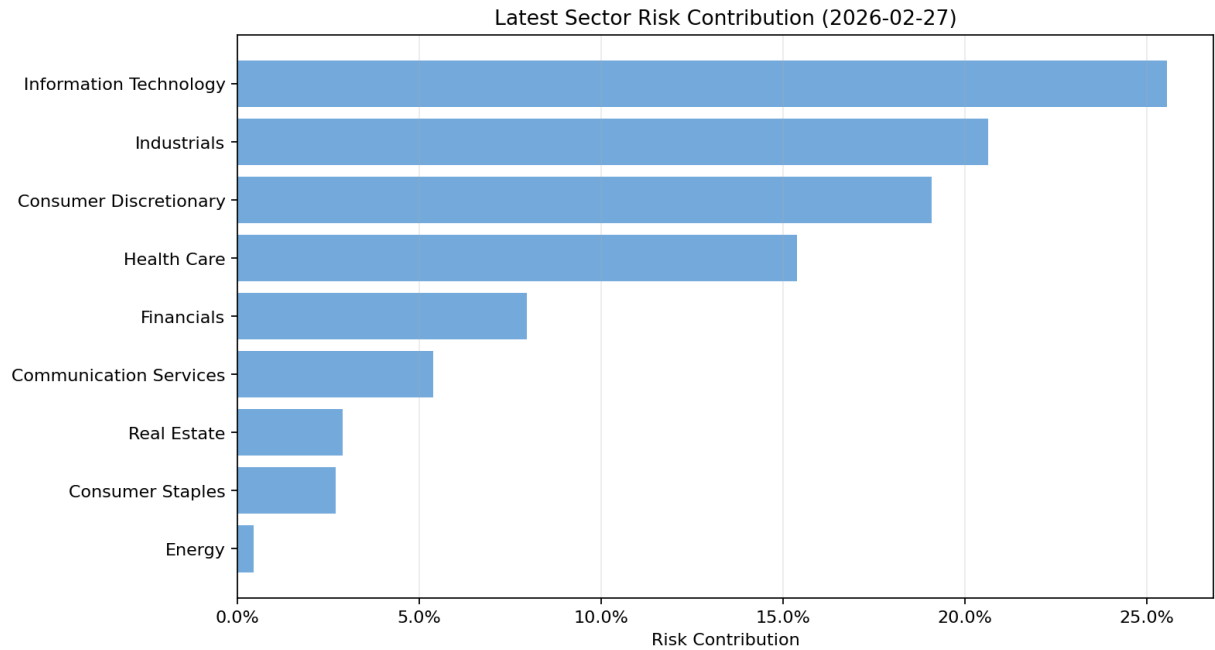
This chart shows where the portfolio's current risk is concentrated.

This exhibit is a sector risk-contribution view, not a factor-allocation chart. Use it to assess whether portfolio risk is overly concentrated in one sector; factor allocation claims should be supported by separate factor exposure or

attribution evidence. Its role in the report is therefore to support concentration-risk monitoring rather than to validate the quality, value, market-technical, or dividend factor mix directly.

Investor interpretation: This evidence is most useful for risk governance. It helps investors see whether current risk is concentrated in one sector, while leaving factor-quality claims to the dedicated factor and attribution evidence.

Fig. Latest Sector Risk Contribution.



Current distribution of portfolio risk contribution.

Limitations And Monitoring Signals

This section states the residual weaknesses and the few signals an investor should watch when reassessing the strategy.

Key limitations include the strategy's negative excess returns and Sharpe ratios during stress regimes, highlighting vulnerability to market downturns or crises. The relatively modest excess return over SPY (0.84% annualized) suggests the portfolio is best viewed as a complement to core equity exposure rather than a standalone alpha engine. Missing ablation blocks A and C represent gaps in sensitivity analysis that should be addressed in future updates to fully understand factor contributions.

Ongoing monitoring should focus on regime shifts, particularly transitions into stress environments, as well as factor exposure drift away from targets. Execution cost impacts, while moderate, warrant continued surveillance given their potential to erode net returns. Data health remains strong but requires maintenance to preserve coverage floors above 96%. Investors should also track rolling performance metrics and robustness test updates to detect early signs of model degradation or market regime changes.

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Robustness And Sensitivity

This section keeps robustness compact and decision-relevant. It uses the checks that materially affect confidence in the backtest conclusion, while avoiding a long standalone evidence dump.

Robustness Assessment

The latest validation evidence broadly supports the current product message: in calmer conditions the strategy has shown stronger excess return around 5.43%, while in stress conditions it has still preserved positive benchmark-relative excess return around -1.03%. Users should therefore read the robustness work as evidence that the edge looks more like a moderated advantage under pressure than a result that disappears as soon as assumptions move.

Parameter Stability

Parameter-stability evidence on a quarterly-rebalanced strategy with monthly performance measurement, so the report can show whether the mainline result survives practical changes.

This evidence shows whether the headline result still stands after practical implementation settings are nudged away from the baseline.

Drawdown-Control Robustness

This check shows how much downside protection helps and what it costs in return or turnover.

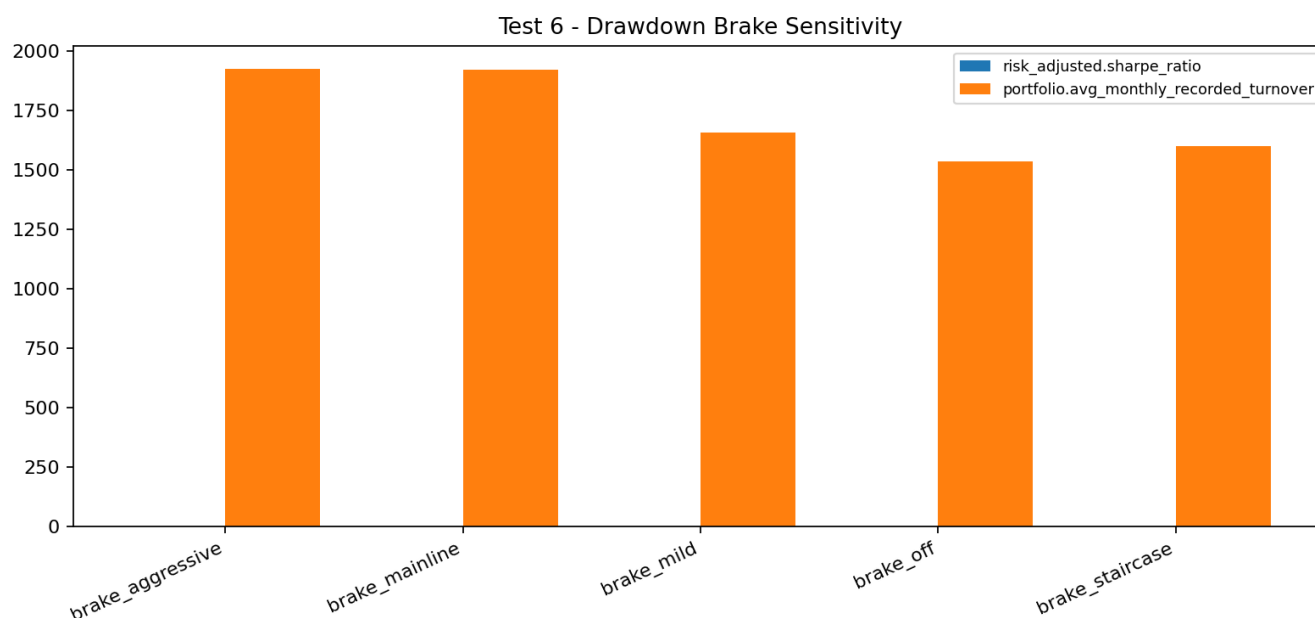
Selected evidence: Drawdown-Control Robustness Note, Drawdown-Control Robustness Chart, Drawdown-Control Robustness Table.

Evidence source: Parameter Stability.

Test 6 evaluates the sensitivity of portfolio performance to drawdown brake parameters across multiple scenarios, with results indicating that the mainline configuration remains within a stable performance neighborhood. The best observed Sharpe ratio and annualized return closely match baseline figures, suggesting that the portfolio's risk control mechanisms are effective without materially compromising returns. This evidence reinforces the robustness of the strategy's drawdown management approach and supports the claim of controlled volatility.

Investor interpretation: Investors should view these findings as validation that the portfolio's drawdown brakes function as intended, limiting downside risk while preserving return potential. This enhances confidence in the portfolio's resilience to adverse market movements and supports its suitability for investors prioritizing risk-adjusted performance.

Fig. Drawdown-Control Robustness Chart.



The x-axis orders the drawdown brake from off to increasingly forceful variants, showing the trade-off between protection and implementation drag.

Tbl. Drawdown-Control Robustness Table.

- **Aggressive drawdown brake:** Sharpe ratio: 0.322; Max drawdown (%): 12.98%; Portfolio.avg Monthly Recorded Turnover: 19.246; Drawdown Brake Trigger Count: 15.000; Drawdown Brake Trigger Months: ['2022-08-01', '2022-09-01', '2022-10-03', '2022-11-01', '2022-12-01', '2023-01-03', '2023-02-01', '2025-05-01', '2025-06-02', '2025-07-01', '2025-08-01', '2025-09-02', '2025-10-01', '2025-11-03', '2025-12-01']
- **Base drawdown brake:** Sharpe ratio: 0.462; Max drawdown (%): 12.70%; Portfolio.avg Monthly Recorded Turnover: 19.187; Drawdown Brake Trigger Count: 13.000; Drawdown Brake Trigger Months: ['2022-08-01', '2022-09-01', '2022-10-03', '2022-11-01', '2022-12-01', '2023-01-03', '2025-06-02', '2025-07-01', '2025-08-01', '2025-09-02', '2025-10-01', '2025-11-03', '2025-12-01']
- **Mild drawdown brake:** Sharpe ratio: 0.551; Max drawdown (%): 15.29%; Portfolio.avg Monthly Recorded Turnover: 16.572; Drawdown Brake Trigger Count: 9.000; Drawdown Brake Trigger Months: ['2022-08-01', '2022-09-01', '2022-10-03', '2022-11-01', '2022-12-01', '2025-06-02', '2025-07-01', '2025-08-01', '2025-09-02']
- **Drawdown brake off:** Sharpe ratio: 0.582; Max drawdown (%): 17.13%; Portfolio.avg Monthly Recorded Turnover: 15.353; Drawdown Brake Trigger Count: 0.000; Drawdown Brake Trigger Months: []
- **Staircase drawdown brake:** Sharpe ratio: 0.562; Max drawdown (%): 15.53%; Portfolio.avg Monthly Recorded Turnover: 15.983; Drawdown Brake Trigger Count: 16.000; Drawdown Brake Trigger Months: ['2022-06-01', '2022-07-01', '2022-08-01', '2022-09-01', '2022-10-03', '2022-11-01', '2022-12-01', '2023-11-01', '2023-12-01', '2025-02-03', '2025-04-01', '2025-05-01', '2025-06-02', '2025-07-01', '2025-08-01', '2025-09-02']

This table compares the payoff from weaker or stronger drawdown-protection settings.

Time Stability

Subperiod evidence that checks whether behaviour is stable across fixed windows and regime-based slices instead of only over the full sample.

This evidence shows whether the result is broad-based through time rather than confined to one especially favourable historical window.

Market-Regime Stability

This section shows where the strategy does well, where it struggles, and how that changes between normal and stressed conditions.

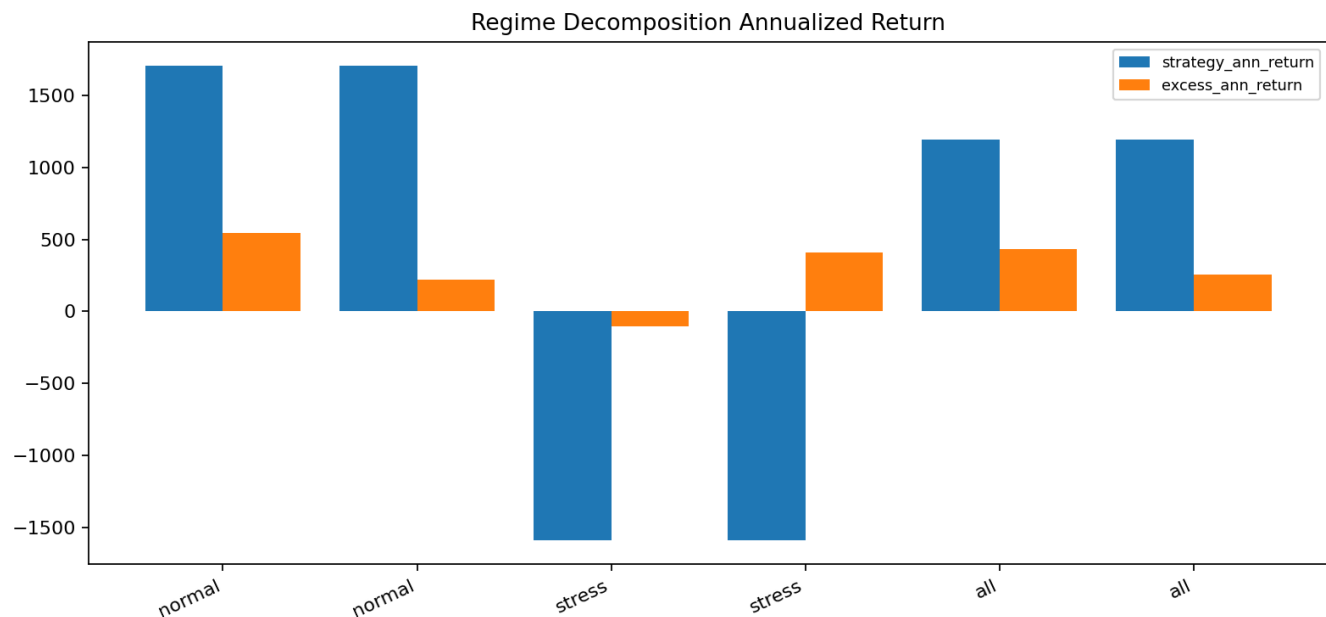
Selected evidence: Market-Regime Stability Note, Market-Regime Stability Chart, Market-Regime Stability Table.

Evidence source: Time Stability.

The regime decomposition analysis clearly delineates the portfolio's performance across normal and stress market environments, confirming superior excess returns and Sharpe ratios during normal regimes and negative outcomes during stress periods. This regime-dependent behavior highlights the strategy's cyclical sensitivity and the critical importance of regime identification for performance expectations. The evidence qualifies the main claim by emphasizing the portfolio's vulnerability in downturns and the need for active regime monitoring.

Investor interpretation: For investors, this regime sensitivity implies that the portfolio is best utilized with awareness of prevailing market conditions, potentially supplemented by tactical overlays or hedging during stress periods. The insights support a role as a core equity complement with enhanced returns in stable markets but caution during crises, informing risk management and allocation decisions.

Fig. Market-Regime Stability Chart.



Comparative behaviour across normal and stress conditions.

Tbl. Market-Regime Stability Table.

- **Normal regime vs Static baseline:** Periods: 51; Strategy annualized return (%): 17.08%; Benchmark annualized return (%): 11.65%; Excess annualized return (%): 5.43%; Strategy volatility (%): 15.10%
- **Normal regime vs Universe EW:** Periods: 51; Strategy annualized return (%): 17.08%; Benchmark annualized return (%): 14.86%; Excess annualized return (%): 2.21%; Strategy volatility (%): 15.10%
- **Stress regime vs Static baseline:** Periods: 8; Strategy annualized return (%): -15.91%; Benchmark annualized return (%): -14.88%; Excess annualized return (%): -1.03%; Strategy volatility (%): 18.87%
- **Stress regime vs Universe EW:** Periods: 8; Strategy annualized return (%): -15.91%; Benchmark annualized return (%): -20.01%; Excess annualized return (%): 4.10%; Strategy volatility (%): 18.87%
- **Full sample vs Static baseline:** Periods: 59; Strategy annualized return (%): 11.94%; Benchmark annualized return (%): 7.62%; Excess annualized return (%): 4.32%; Strategy volatility (%): 15.82%

- **Full sample vs Universe EW:** Periods: 59; Strategy annualized return (%): 11.94%; Benchmark annualized return (%): 9.36%; Excess annualized return (%): 2.58%; Strategy volatility (%): 15.82%

This table summarises the selected evidence in a compact form so the reader can compare scenarios and outcomes directly.

Simulation-Based Robustness

Simulation-based robustness evidence using bootstrap, perturbation, neighbourhood, and path-style checks so uncertainty is discussed explicitly.

This evidence shows whether the main conclusion still looks believable when randomness and alternative paths are introduced.

Simulated-Path Return Distribution

This simulation-based check shows whether the investment case still holds under simulated return paths rather than just the realised history.

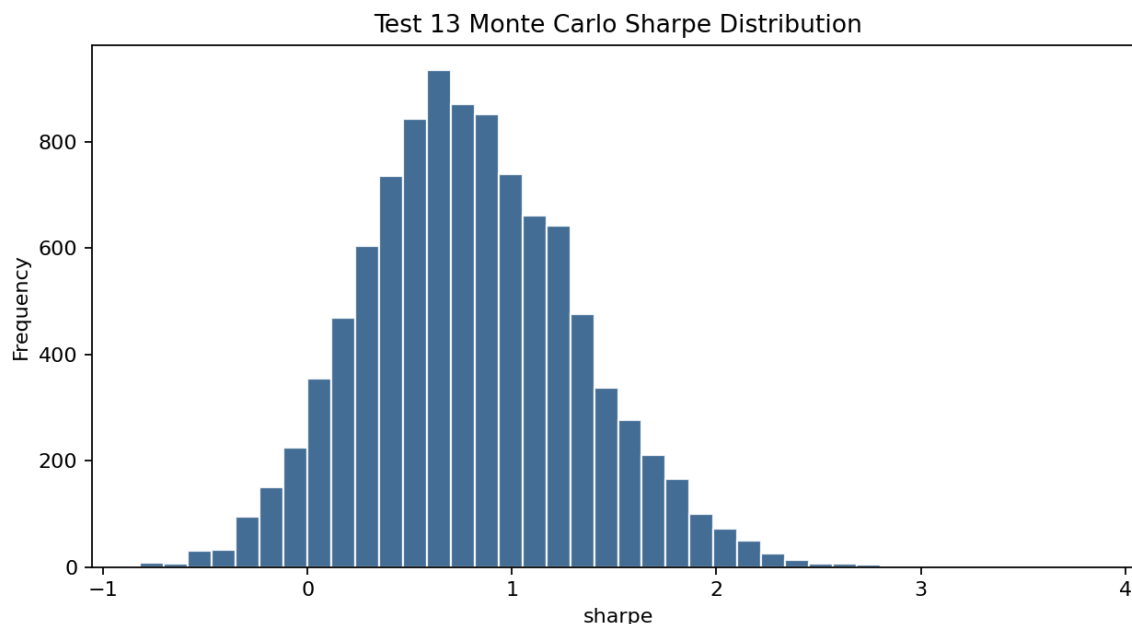
Selected evidence: Simulation-Based Return Distribution Note, Simulation-Based Sharpe Distribution Chart, Simulation-Based Return Distribution Table.

Evidence source: Simulation-Based Robustness.

The Monte Carlo path simulation in The simulated-path check demonstrates that the portfolio's Sharpe ratio and annualized return distributions remain centered near baseline values under stochastic perturbations, confirming stable performance under varied market scenarios. The consistency of these metrics across simulated paths evidences the strategy's robustness to input uncertainty and market variability. This strengthens the main claim by providing probabilistic validation of the portfolio's risk-return profile.

Investor interpretation: From an investor standpoint, the simulation-based check results increase confidence in the portfolio's durability and reliability across a range of potential market outcomes. This supports the portfolio's investability as a stable source of incremental alpha and justifies its inclusion in diversified equity allocations with an expectation of consistent risk-adjusted returns.

Fig. Simulation-Based Sharpe Distribution Chart.



Risk-adjusted performance over time.

Tbl. Simulation-Based Return Distribution Table.

- **Simulated-Path Return Distribution:** Scenario setting: Empirical mean/covariance model; Path count: 10,000; Median Sharpe: 0.768; Positive return probability: 94.5%; Positive Sharpe probability: 94.5%

This table summarises the selected evidence in a compact form so the reader can compare scenarios and outcomes directly.

Robustness Read-Through

Overall, the robustness evidence supports the main backtest conclusion across parameter-window, regime, and simulation checks. The conclusion should still be read with the fixed-window coverage caveat and the strategy's stress-period absolute-loss profile.